Math Inquirers

Extension Workbook

with answers

Level 6

Self-study | Ages 11-13



My name _____

About this book

Math Inquirers Extension Workbook, Level 6
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
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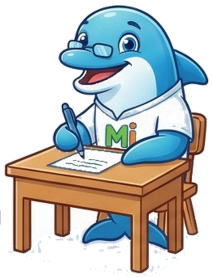
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Code the pattern	Not in this proof edition				☐
2	Prime hunters	Not in this proof edition				☐
3	Below zero	Not in this proof edition				☐
4	Algebra machines	Not in this proof edition				☐
5	Every possible way	Not in this proof edition				☐
6	Fair or unfair?	Not in this proof edition				☐
7	Turtle geometry	6-7	8-9	10 and 12	11	☐
8	Secret codes	Not in this proof edition				☐
9	Data detectives	Not in this proof edition				☐
10	Showcase challenge	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

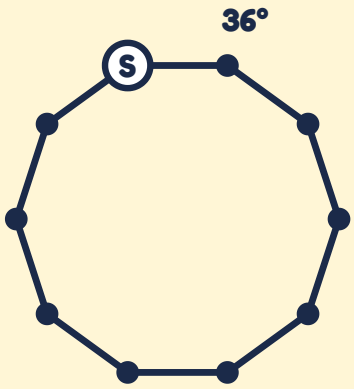
Learn it: Angles in polygons

My goal. I can find the turn angle and the inside angle of a regular polygon. I can predict what a turtle program draws.

A turtle moves forward and turns. Going once round a closed shape, it turns 360° in all and faces the way it began. In a regular polygon with n sides every turn is 360° ÷ n, and the inside angle is 180° – the turn.

Worked example

A turtle draws a regular polygon with 10 sides. Find the turn at each corner and the inside angle.



1. **Going once round, the turtle turns 360° in all.**
2. **There are 10 equal turns: 360° ÷ 10 = 36° at each corner.**
3. **The turn and the inside angle make a straight line: 180° – 36° = 144°.**
4. **In code: repeat 10 [move 60, turn ↻ 36].**

Answer: The turn is 36° and the inside angle is 144°.

How to find the angles of a regular polygon

1. Going once round, the turtle turns 360° in all.
2. Turn = 360° ÷ number of sides.
3. Inside angle = 180° – turn.
4. Test it in code: repeat (sides) [move, turn].

Your turn

Answer at the back of the book.

Try A regular polygon has 18 sides. The turn at each corner is ☐ ° and the inside angle is ☐ °.

Start here

Do these first. Take your time.

1 **Complete the table for regular polygons.**

Sides	Turn	Inside angle
5		
12		
	18°	
		172°

2 **Program A: repeat 10 [move 40, turn ↻ 36]. Program B: repeat 12 [move 40, turn ↻ 30]. Program C: repeat 6 [move 40, turn ↻ 30]. One of them draws a regular polygon with inside angles of 150°.**

(a) The program is A B C

(b) Program C turns ☐ ° in all, so its path closes does not close

3 **Trace the code. Write the total turned after each repeat, and the turn from the start direction (the total with every full 360° taken away).**



Repeat	Total	From start
1		
2		
3		
4		
5		

(a) After repeat 5 the turn from the start is ☐ °.

(b) Is the turtle facing the way it started? Yes No

Start here

Keep going. Check your work.

4 **A turtle walks round a closed shape with 5 sides. Its first 4 turns are 60°, 85°, 110° and 50°. It faces the way it started when it is back.**

(a) The last turn is ☐°. (b) The inside angle there is ☐°.

(c) The 5 inside angles add up to ☐°.

Working: _____

5 **A regular polygon has 18 sides. Find its angles.**

(a) The turn at each corner is ☐°.

(b) Each inside angle is ☐°.

(c) All the inside angles add up to ☐°.

Working: _____

6 **Circle each turn angle that draws a regular polygon. The number of sides must be a whole number.**

24°	30°	45°	50°	64°	72°	100°	120°	150°

7 **A turtle draws a star with 10 points and winds round the centre 3 times. It runs repeat 10 [move 80, turn ↻ ?].**

(a) The total turning is ☐°.

(b) Each turn is ☐°.

(c) The angle at each point is ☐°.

Working: _____

Start here

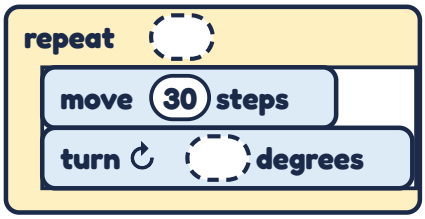
Nearly there. You can do it.

8 **A turtle draws a regular shape with 30 sides. Fill in the blanks.**

(a) repeat ☐

(b) turn ☐ degrees

(c) Each inside angle is ☐°.

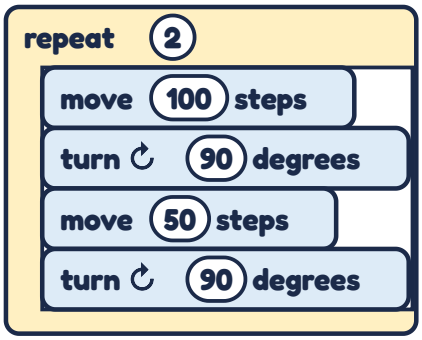


9 **This code draws a rectangle that is not a square. Predict it.**

(a) The turtle turns ☐° in all.

(b) It walks ☐ steps in all.

(c) Change 50 to 80. It walks ☐ steps.



10 **Shapes fit around a point when their inside angles add to 360°. Check each group.**

(a) 6 triangles: ☐°

(b) 3 squares: ☐°. Is there a gap? Yes No

(c) 2 hexagons and 2 triangles: ☐°

11 **Each inside angle is a few times the turn angle. Find the number of sides.**

(a) 2 times as big: ☐ sides. (b) 5 times as big: ☐ sides.

(c) 8 times as big: ☐ sides.

Working: _____

Think harder

Hints are upside down. Try first!

1 **A turtle repeats [move 40, turn ↻ 100] again and again.**

(a) After 18 repeats it has turned ☐ ° in all.

(b) That is ☐ full turns.

(c) It first faces the start direction after ☐ repeats.

Working: _____

_____

Hint. Find the first multiple of 100 that is also a multiple of 360.

2 **All the inside angles of a regular polygon total 2340°.**

(a) The polygon has ☐ sides. **(b)** Each inside angle is ☐ °.

(c) Each turn is ☐ °.

Working: _____

_____

Hint. A shape with n sides has (n − 2) × 180 in all.

3 **A triangle, 2 squares and one more regular polygon fit around a point with no gap. All the sides are the same length.**

(a) The inside angle of the fourth shape is ☐ °.

(b) It has ☐ sides.

Working: _____

_____

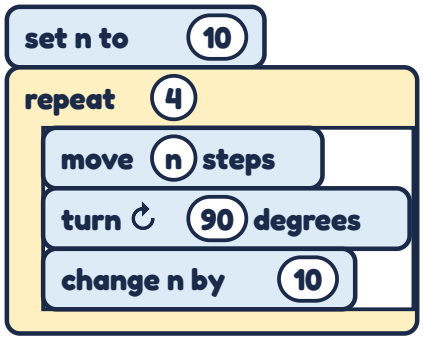
Hint. The angles must add to 360. What is left for the fourth?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 6: Where does it end?

The turtle starts at x = 0, y = 0, facing right.



(a) After 4 repeats: x = ☐ , y = ☐

(b) After 12 repeats: x = ☐ , y = ☐

Hint. Make a table. Write x and y after each repeat.

2 Beyond Level 6: Points of a star.

The angle at each point of a star is 180° – the turn. Add up all the point angles.

(a) 5-point star (turn 144°): the 5 point angles add to ☐ °.

(b) 8-point star, 3 times round: the 8 point angles add to ☐ °.

(c) 10-point star, 3 times round: each point angle is ☐ ° and they add to ☐ °.

Hint. Find each turn first. Then subtract it from 180 and multiply.

3 Beyond Level 6: A polygon that looks like a circle.

The turtle runs repeat 360 [move 1, turn ↻ 1].

(a) The path is ☐ steps long.

(b) The distance across the shape is about ☐ steps (use 3.14, nearest whole number).

Hint. The path round the shape is about 3.14 times the distance across.

✔ Check yourself

No hints. Try each one on your own.

1 A regular polygon has 40 sides. Each turn is ☐° and each inside angle is ☐°.

Working: _____

2 A turtle turns 45° at each corner of a regular polygon. The polygon has ☐ sides.

Working: _____

3 A regular polygon has inside angles of 156° each. It has ☐ sides.

Working: _____

4 A turtle goes round a closed 4-sided shape. Its first 3 turns are 95°, 80° and 105°. The last turn is ☐°.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Code the pattern	☐	☐	☐	☐	/ 4	
2	Prime hunters	☐	☐	☐	☐	/ 4	
3	Below zero	☐	☐	☐	☐	/ 4	
4	Algebra machines	☐	☐	☐	☐	/ 4	
5	Every possible way	☐	☐	☐	☐	/ 4	
6	Fair or unfair?	☐	☐	☐	☐	/ 4	
7	Turtle geometry	☐	☐	☐	☐	/ 4	
8	Secret codes	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Showcase challenge	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 6.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 7 · Turtle geometry

Your turn
page 6

Answer: 20°, 160°
Why: 360° ÷ 18 = 20° for the turn. The inside angle is 180° − 20° = 160°.

START HERE

1
page 7

Answer: Row 1: 72°, 108°. Row 2: 30°, 150°. Row 3: 20 sides, 162°. Row 4: 45 sides, 8°.
Why: Turn = 360 ÷ sides and inside angle = 180 − turn. 5 sides: turn 72, inside 108. 12 sides: turn 30, inside 150. Turn 18: 360 ÷ 18 = 20 sides, inside 162. Inside 172: turn 8, and 360 ÷ 8 = 45 sides.

2
page 7

Answer: (a) B (b) 180°, does not close
Why: Inside 150° means a turn of 180° − 150° = 30°, and 360° ÷ 30° = 12 sides: Program B. Program C also turns 30°, but only 6 × 30 = 180° in all. That is half a turn, so the path does not close.

3
page 7

Answer: (a) 140 (b) No
Why: The total turned grows by 100 each repeat: 100, 200, 300, 400, 500. Take away 360 when the total passes it: 500 − 360 = 140. The turtle faces 140° from its start direction, not 0°, so it is not facing the way it began.

4
page 8

Answer: (a) 55 (b) 125 (c) 540
Why: All the turns add to 360°. The first 4 add to 60 + 85 + 110 + 50 = 305, so the last turn is 360 − 305 = 55°. The inside angle there is 180 − 55 = 125°. The 5 inside angles add to 5 × 180 − 360 = 540°.

5
page 8

Answer: (a) 20 (b) 160 (c) 2880
Why: Turn: 360 ÷ 18 = 20°. Inside angle: 180 − 20 = 160°. All 18 inside angles: 18 × 160 = 2880°. Check with (18 − 2) × 180 = 16 × 180 = 2880.

6
page 8

Answer: 24°, 30°, 45°, 72°, 120°
Why: The turn must divide 360 exactly. 360 ÷ 24 = 15 sides, ÷ 30 = 12, ÷ 45 = 8, ÷ 72 = 5 and ÷ 120 = 3 sides. 360 ÷ 50, 64, 100 and 150 are not whole numbers.

7
page 8

Answer: (a) 1080 (b) 108 (c) 72
Why: Going round 3 times, the turtle turns 3 × 360 = 1080° in all. 1080 ÷ 10 = 108° at each point. The angle at each point is 180 − 108 = 72°.

8
page 9

Answer: (a) 30 (b) 12 (c) 168
Why: 30 sides need 30 repeats. The turn is 360 ÷ 30 = 12°. The inside angle is 180 − 12 = 168°.

9
page 9

Answer: (a) 360 (b) 300 (c) 360
Why: The rectangle has 4 right-angle turns: 4 × 90 = 360°. The turtle walks 100 + 50 + 100 + 50 = 300 steps. With 80 instead of 50: 100 + 80 + 100 + 80 = 360 steps.

10
page 9

Answer: (a) 360 (b) 270°, Yes (c) 360
Why: Triangle angle 60: 6 × 60 = 360. Square angle 90: 3 × 90 = 270, which is less than 360, so there is a gap of 90. Hexagon angle 120: 2 × 120 + 2 × 60 = 360.

11
page 9

Answer: (a) 6 (b) 12 (c) 18
Why: Turn + inside = 180. If the inside angle is k times the turn, then (k + 1) × turn = 180. k = 2: turn 60, so 6 sides. k = 5: turn 30, so 12 sides. k = 8: turn 20, so 18 sides.

THINK HARDER

1
page 10

Answer: (a) 1800 (b) 5 (c) 18
Why: After k repeats the total turned is 100k. It faces the start direction when 100k is a multiple of 360: 100k = 1800 first, at k = 18 (1800 ÷ 100). 1800 = 5 × 360, so it has turned round 5 times and draws an 18-point star.

2
page 10

Answer: (a) 15 (b) 156 (c) 24
Why: The sum of the inside angles is (n − 2) × 180, so n − 2 = 2340 ÷ 180 = 13 and n = 15. Each inside angle is 2340 ÷ 15 = 156°. The turn is 180 − 156 = 24°. Check: 360 ÷ 24 = 15.

3
page 10

Answer: (a) 120 (b) 6
Why: The angles are 60 + 90 + 90 = 240. 360 − 240 = 120° is left. A regular polygon with inside angle 120° has turn 60°, so 360 ÷ 60 = 6 sides: a hexagon.

GO FURTHER

1
page 11

Answer: (a) −20, 20 (b) −60, 60
Why: Right 10, down 20, left 30, up 40 gives x = 10 − 30 = −20 and y = −20 + 40 = 20 after 4 repeats. The next 4 moves are right 50, down 60, left 70, up 80: x = −20 + 50 − 70 = −40 and y = 20 − 60 + 80 = 40. Every 4 repeats adds −20 to x and 20 to y, so after 12 repeats it is at (−60, 60).

2
page 11

Answer: (a) 180 (b) 360 (c) 72°, 720°
Why: 5-point star: turn 144, point angle 36, and 5 × 36 = 180. 8-point star going round 3 times: turn 135, point angle 45, and 8 × 45 = 360. 10-point star going round 3 times: turn 108, point angle 72, and 10 × 72 = 720.

3
page 11

Answer: (a) 360 (b) 115
Why: 360 turns of 1° add to 360°, one full turn, so the path closes. Inside angle: 180 − 1 = 179°. The path is 360 steps long. For a circle, the distance round is about 3.14 × the distance across, so the distance across is about 360 ÷ 3.14 = 114.6, which is 115 to the nearest whole number.

CHECK YOURSELF

1
page 12

Answer: 9°, 171°
Why: 360 ÷ 40 = 9°. The inside angle is 180 − 9 = 171°.

2
page 12

Answer: 8
Why: 360 ÷ 45 = 8 sides.

3
page 12

Answer: 15
Why: The turn is 180 − 156 = 24°, and 360 ÷ 24 = 15 sides.

4
page 12

Answer: 80°
Why: The turns add to 360. 95 + 80 + 105 = 280, so the last turn is 360 − 280 = 80°.